

## Project Title

# Regularized Multivariate Meta - Modelling for Improved Mortality Prediction in Cardiac Arrest ICU Patients

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**Project Guide**

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## **ABSTRACT**

Accurate prediction of in-hospital mortality among intensive care unit (ICU) patients following cardiac arrest is crucial for timely clinical decision-making and efficient resource allocation. This project focuses on developing a predictive framework that integrates both structured clinical data and unstructured textual information to improve mortality prediction.

The study utilizes the MIMIC-IV dataset, which includes patient demographics, vital signs, laboratory measurements, comorbidities, and treatment indicators, along with clinical notes such as discharge summaries and radiology reports. Structured data were pre-processed through cleaning, imputation, and feature engineering, while textual data were transformed using techniques such as TF-IDF and BioBERT embedding's. Dimensionality reduction methods like SVD and PCA were applied to enhance computational efficiency.

Feature selection was performed using LASSO and XGBoost, followed by the development of interpretable logistic regression models. The performance of models based solely on structured data was compared with multimodal models that incorporated textual features. Evaluation metrics such as AUC, accuracy, F1-score, and recall were used to assess model effectiveness.

The results demonstrate that incorporating unstructured clinical text significantly improves predictive performance. The multimodal model achieved an AUC of approximately 0.92 compared to 0.75 for structured data alone, highlighting the benefit of textual information in capturing clinical context.

Overall, this project shows that combining structured and unstructured data enhances both prediction accuracy and clinical interpretability, providing a robust and practical approach for mortality risk prediction in ICU settings.

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## **INTRODUCTION**

Cardiac arrest is one of the most critical medical emergencies worldwide and is associated with high mortality rates, especially among patients admitted to intensive care units (ICUs). Despite advancements in medical technology and critical care practices, predicting patient outcomes remains a significant challenge. Early and accurate prediction of in-hospital mortality is essential for timely clinical interventions, improved patient management, and efficient utilization of healthcare resources.

Traditionally, mortality prediction models in ICUs have relied on structured clinical data such as vital signs, laboratory results, demographic information, and comorbidities. While these models provide valuable insights, they often overlook unstructured clinical information, including discharge summaries and radiology reports. Such textual data contain rich contextual details about patient conditions, clinical reasoning, and treatment progression, which can significantly enhance predictive performance.

This project focuses on developing an advanced predictive framework that integrates both structured and unstructured data using the MIMIC-IV dataset. The study employs statistical and machine learning techniques such as logistic regression, LASSO (Least Absolute Shrinkage and Selection Operator), and XGBoost for feature selection and modelling. Additionally, natural language processing methods like TF-IDF and BioBERT embeddings are used to extract meaningful features from clinical text.

The main objective of this project is to improve the accuracy and interpretability of mortality prediction models by combining multimodal data sources. By comparing structured-only models with multimodal models, the study aims to demonstrate the added value of textual information in enhancing predictive performance.

Overall, this project contributes to the growing field of healthcare analytics by presenting an interpretable and effective approach for mortality prediction in ICU patients. The findings highlight the importance of integrating diverse data sources to support better clinical decision-making and improve patient outcomes.

## **MOTIVATION**

Cardiac arrest remains a major cause of mortality worldwide, particularly among patients admitted to intensive care units (ICUs). Despite the availability of advanced monitoring systems and clinical expertise, accurately predicting patient outcomes at an early stage continues to be a difficult task. This creates a critical need for more reliable and data-driven approaches to support clinical decision-making.

Most existing prediction models rely primarily on structured clinical data such as vital signs, laboratory results, and patient demographics. While these variables provide important physiological information, they often fail to capture the full clinical picture. Unstructured data, such as discharge summaries and radiology reports, contain valuable insights into patient condition, disease progression, and physician judgment, yet they are often underutilized in traditional models.

The motivation behind this project is to bridge this gap by integrating both structured and unstructured data to enhance mortality prediction accuracy. By leveraging machine learning techniques and natural language processing methods, this study aims to extract meaningful patterns from complex healthcare data.

Another key motivation is to develop models that are not only accurate but also interpretable. In healthcare, it is essential for predictive models to be transparent and understandable so that clinicians can trust and effectively use them in real-world settings.

Ultimately, this project is driven by the goal of improving patient outcomes, enabling early identification of high-risk patients, and supporting better allocation of medical resources in ICUs. It highlights the importance of combining multiple data sources to create more powerful and clinically useful predictive systems.

## **AIM & OBJECTIVES**

### **Aim**

The primary objective of this study is to develop intelligent, data-driven models to accurately predict mortality risk among ICU patients following cardiac arrest. The aim is to enable early identification of high-risk patients, thereby supporting timely clinical intervention and improved patient outcomes.

### **1. Data Integration**

To integrate heterogeneous data from multiple modules of the MIMIC-IV database, including ICU records, hospital data, radiology reports, and discharge summaries, into a unified and structured dataset suitable for comprehensive analysis.

### **2. Model Development**

To develop a robust machine learning framework that combines structured clinical variables with unstructured textual data (such as clinical notes and radiology reports) in order to enhance the accuracy of mortality prediction.

### **3. Performance Evaluation**

To evaluate the predictive performance and clinical usefulness of the developed models by assessing the impact of incorporating unstructured clinical text alongside structured data, using appropriate statistical and machine learning evaluation metrics.

# **DATASET**

## **Source**

The dataset used in this study is the MIMIC-IV (Medical Information Mart for Intensive Care IV) database, publicly available at [www.physionet.org](http://www.physionet.org). MIMIC-IV is a large, freely accessible clinical database developed by the MIT Laboratory for Computational Physiology. It contains de-identified health-related data of patients admitted to the Beth Israel Deaconess Medical Center (BIDMC) in Boston, Massachusetts.

## **Data Components**

The MIMIC-IV database consists of two broad categories of data used in this study:

1. **Structured Data:** Sourced from the MIMIC-IV Database, including the Hospital Module (containing patient demographics, diagnoses, laboratory results, and medication orders) and the ICU Module (containing time-series records of vital signs, procedures, and ICU-specific measurements such as GCS scores and ventilation parameters).
2. **Unstructured Data:** Sourced from MIMIC-IV Notes, comprising clinical free-text documents of two types: Discharge Summaries (comprehensive narratives written by physicians at the time of patient discharge) and Radiology Reports (structured and semi-structured text reports from radiological examinations).



## **Patient Cohort Selection**

Starting from over 364,000 unique patients in the full MIMIC-IV database, cardiac arrest patients were identified using ICD-9 and ICD-10 diagnostic codes, yielding 2,653 candidate patients. After applying inclusion criteria, first ICU stay only and age 18 years or older — the final study cohort comprised 2,307 patients. Of these, 1,207 patients (52.3%) died during their hospital stay (mortality = 1) and 1,100 patients (47.7%) survived to discharge (mortality = 0).

## **DATASET DESCRIPTION**

The dataset integrates multiple clinical domains to provide a comprehensive view of each patient. The variables are organized into five predictor categories and one target variable:

### **Vital Signs**

Six continuous physiological measurements recorded during ICU stay: Heart Rate (HR), Systolic Blood Pressure (SBP), Diastolic Blood Pressure (DBP), Respiratory Rate (RR), Oxygen Saturation (SpO2), and Body Temperature (BT). For each variable, the mean, minimum, and maximum values over the first 24 hours were extracted and included as features.

### **Laboratory Tests**

Seven key biochemical markers: Lactate, pH, Creatinine, Blood Urea Nitrogen (BUN), Glucose, Hemoglobin, and White Blood Cell count (WBC). These laboratory values are critical indicators of organ function, metabolic status, and immune response in critically ill patients.

### **Comorbidities**

Four binary indicators representing the presence or absence of pre-existing conditions: Hypertension, Heart Failure, Diabetes, and Chronic Obstructive Pulmonary Disease (COPD). These were extracted from ICD code-based diagnosis records.

### **Treatment Indicators**

Three binary flags indicating whether specific interventions were administered: Mechanical Ventilation, Epinephrine administration, and Dopamine administration. These reflect the severity of the patient's clinical condition and the intensity of care required.

### **Neurological Assessment**

The Glasgow Coma Scale (GCS) components Eye response, Verbal response, and Motor response and the total GCS score. These are among the most clinically important predictors of neurological recovery following cardiac arrest.

## **Target Variable**

In-hospital mortality, coded as a binary variable: 1 (Died) or 0 (Survived). This is the outcome to be predicted by all models developed in this study.

# DATA PRE-PROCESSING

## Structured Data Pre-processing

### 4.1 Data Sources

Structured data were extracted from the MIMIC-IV database, including ICU records such as vital signs, laboratory measurements, comorbidities, treatment interventions, and neurological assessments. These data provide comprehensive clinical information for each patient admission.

### 4.2 Cohort Selection

The study cohort was constructed by identifying cardiac arrest patients using ICD-9 and ICD-10 codes. Inclusion criteria were restricted to adult patients (age  $\geq 18$ ) and their first ICU stay. Formally, the study population can be defined as:

$$\mathcal{D} = \{i : \text{ICU patients with cardiac arrest, age} \geq 18\}$$

After applying these criteria, a total of 2,307 patients were included in the final analysis.

### 4.3 Feature Extraction

Clinical features were extracted across multiple domains:

- **Vital Signs:** heart rate, blood pressure, respiratory rate, SpO<sub>2</sub>, temperature
- **Laboratory Tests:** lactate, pH, creatinine, BUN, glucose, hemoglobin, WBC, electrolytes
- **Comorbidities:** hypertension, diabetes, COPD, heart failure
- **Treatment Indicators:** mechanical ventilation, epinephrine, dopamine
- **Neurological Assessment:** Glasgow Coma Scale (GCS)

### 4.4 Data Cleaning

Data cleaning involved removing duplicate records, filtering invalid or zero values, and ensuring consistency across timestamps. Observations were retained only if they fell within physiologically plausible ranges:

$$X_{clean} = \{x_i \in X : x_i \text{ is valid and within acceptable range}\}$$

Additionally, all clinical measurements were restricted to the first 24 hours of ICU admission to maintain temporal consistency.

## 4.5 Feature Engineering

To transform time-series clinical data into structured features, statistical aggregations were computed for each variable within the first 24 hours.

The **mean** value was calculated as:

$$\bar{x}_i = \frac{1}{n} \sum_{t=1}^n x_{it}$$

The **minimum** and **maximum** values were computed as:

$$x_i^{min} = \min(x_{i1}, x_{i2}, \dots, x_{in})$$

$$x_i^{max} = \max(x_{i1}, x_{i2}, \dots, x_{in})$$

Binary indicators were created for treatments and comorbidities:

$$Z_i = \begin{cases} 1, & \text{if condition/treatment present} \\ 0, & \text{otherwise} \end{cases}$$

The Glasgow Coma Scale (GCS) total score was computed as:

$$GCS_{total} = GCS_{eye} + GCS_{verbal} + GCS_{motor}$$

These transformations convert irregular time-series data into fixed-length, interpretable features suitable for modeling.

## 4.6 Missing Value Handling

Missing values were handled using **Multiple Imputation by Chained Equations (MICE)** using the IterativeImputer from scikit-learn (Pe dregosa et al. 2011) with  $m = 5$  iterations, where each variable is iteratively modeled conditional on other variables:

$$X_j^{(m)} \sim P(X_j | X_{-j})$$

Parameter estimates were combined using Rubin's Rules (Lee and Simpson 2014) to account for within- and between-imputation variance:

$$\hat{\beta}_{\text{MI}} = \frac{1}{m} \sum_{i=1}^m \hat{\beta}_i, \quad T = V + \left(1 + \frac{1}{m}\right) B$$

The resulting complete dataset retained all 2,307 patients without casewise deletion, ensuring consistency for subsequent modeling and analysis. This approach preserves relationships among variables and provides more reliable estimates compared to simple imputation methods.

## 4.7 Final Structured Dataset

The final structured dataset consists of aggregated clinical features, binary indicators, and demographic variables, represented as:

$$X \in \mathbb{R}^{n \times p}$$

where:

- nnn = number of patients
- ppp = number of features

The target variable is defined as:

$$Y = \begin{cases} 1, & \text{if patient died} \\ 0, & \text{otherwise} \end{cases}$$

This processed dataset serves as the input for subsequent modeling, ensuring clinical relevance, statistical robustness, and interpretability.

These preprocessing steps ensure that raw clinical data are transformed into structured, meaningful features suitable for predictive modeling.

### Unstructured Data Pre-processing

#### Stage 1: Text Source Selection & Cohort Filtering

The first stage establishes which clinical documents are eligible for inclusion. The raw MIMIC-IV-Note extension contains over 2.6 million de-identified clinical notes across the entire hospital system — roughly 2.3 million radiology reports and 330,000 discharge summaries, covering 230,000+ patients. This corpus is far broader than the study's analytical cohort and must be precisely filtered before any linguistic processing begins.

All notes are cross-referenced against the structured cohort using the shared hospital admission identifier `hadm_id`. Only notes associated with the 2,307 ICU patients who experienced cardiac arrest and survived initial admission filtering are retained. This alignment is non-trivial: it prevents data leakage from unrelated admissions and ensures temporal consistency with the structured features (which are all drawn from the same admissions).

For **discharge summaries**, one summary per admission is retained when available, yielding 1,618 documents (70.1% cohort coverage). For **radiology reports** — which often have multiple entries per admission (repeat chest X-rays, for instance) — only the earliest report per stay is selected. The rationale here is temporal relevance: the first radiology report most closely mirrors the early ICU window from which structured vitals and labs were extracted. This yields 1,639 radiology notes (71.0% coverage). The result is a maximum of one text document per modality per patient, preserving a strict one-to-one relationship with the structured dataset rows.

## **Stage 2: Text Normalization & Dual-Pipeline Feature Extraction**

Once the eligible documents are isolated, they enter two parallel and complementary feature extraction pipelines. The dual-pipeline design is deliberate: TF-IDF captures *statistical lexical patterns* specific to the corpus, while BioBERT captures *semantic and contextual meaning* drawn from biomedical pretraining. Together, they provide complementary signal.

**Pipeline A — TF-IDF Representation:** Each document is first normalized through classical NLP preprocessing: text is lowercased, punctuation and digits are stripped, and English stopwords (common, low-information words like "the", "is", "of") are removed. A 500-term vocabulary is then constructed *separately* for discharge summaries and radiology reports, because these document types have structurally and lexically distinct content — a discharge summary discusses clinical course and prognoses, while a radiology report follows standardized sections (indication, findings, impression). Treating them with separate

vocabularies respects their linguistic register differences. Sparse document-term matrices are then generated using TF-IDF scoring, which weights terms by their frequency in a document relative to their rarity across all documents, rewarding informative terminology.

### **Pipeline B — BioBERT Embeddings (Detailed Mechanism):**

The study uses BioBERT (specifically the publicly available ClinicalBERT variant by Alsentzer et al. 2019), a transformer model built on the BERT architecture and pretrained on large-scale biomedical text corpora including PubMed abstracts and full-text PubMed Central articles.

Each clinical note is passed through BioBERT's tokenizer, which converts the raw text into subword tokens using a WordPiece vocabulary tuned to biomedical language (e.g., "myocardial" is kept intact rather than split into meaningless fragments). The tokenized sequence is then fed through BioBERT's 12-layer transformer stack. At each layer, every token's representation is refined via **multi-head self-attention** — a mechanism that lets the model dynamically weight the relevance of every other token in the sequence when encoding a given token. This is what gives BERT-style models their power: "arrest" in "cardiac arrest" is encoded differently from "arrest" in "sinus arrest" because the surrounding context is incorporated at every layer.

After the final (12th) transformer layer, each input token has a 768-dimensional dense hidden state vector. To produce a single fixed-length document embedding, the study applies **mean-pooling** across all token hidden states in the final layer. Mathematically: if a document tokenizes to  $N$  tokens, the document embedding is the elementwise average of all  $N$  vectors, yielding one 768-dimensional vector per document. This vector encodes the document's full biomedical semantic content in a continuous, dense representation — far richer than any sparse bag-of-words approach.

BioBERT, a transformer-based model pre-trained on biomedical literature (PubMed and PubMed Central), was used to generate dense semantic embeddings. Each clinical note was tokenized (up to 512 tokens) and passed through the BioBERT model to produce a 768-dimensional embedding vector via mean-pooling of the token representations. The embedding statistics showed a mean of approximately -0.005 and standard deviation of 0.349. Principal Component Analysis (PCA) was subsequently applied for dimensionality reduction, with 107



components retaining 90% of cumulative variance for discharge summaries and 106 components for radiology reports.

### **Stage 3: Dimensionality Reduction — Compressing Without Losing Meaning**

At this point the pipeline has produced high-dimensional feature matrices. TF-IDF matrices are extremely sparse (most terms appear in very few documents), and BioBERT vectors are 768-dimensional dense vectors — both problematic for downstream regression models that suffer from the curse of dimensionality. Stage 3 applies distinct but principled compression strategies to each.

**For TF-IDF features — Truncated SVD:** Singular Value Decomposition (SVD) is applied to decompose the sparse document-term matrix into lower-dimensional latent components that capture the most variance in the data. Unlike standard PCA, Truncated SVD operates directly on sparse matrices without centering, which is essential for TF-IDF data. The number of components retained is chosen to preserve **80% of total variance**. This yields 136 SVD components for discharge summaries and 198 components for radiology reports — a dramatic reduction from the original 500-term vocabularies while retaining the majority of discriminative signal. This technique is also known as Latent Semantic Analysis (LSA) and implicitly captures co-occurrence patterns: words that appear in similar clinical contexts are grouped into shared latent dimensions.

**For BioBERT embeddings — PCA:** Principal Component Analysis is applied to the dense 768-dimensional embedding matrices to retain **90% of explained variance**. This results in 113 principal components for discharge embeddings and 115 for radiology embeddings. The higher variance threshold (90% vs 80%) for BioBERT is justified because these embeddings are information-dense from the start — even small components carry meaningful semantic signal — whereas TF-IDF dimensions are noisier and more redundant at the margins.

### **Stage 4: Integration with Structured Data & Missing Modality Handling**

The reduced text feature matrices — spanning 562 columns in total across both document types and both encoding methods — are merged with the structured clinical dataset using `hadm_id` as the join key. This creates a unified multimodal feature matrix where each row corresponds to a single ICU patient and columns span physiological measurements, lab results, comorbidities, and text-derived embeddings.

A critical design choice here is how to handle patients for whom one or both text modalities are absent. Of the 2,307 patients, 689 lacked discharge summaries, 668 lacked radiology notes, and 712 had at least one missing modality. The study replaces missing text embeddings with zero vectors rather than mean imputation or row deletion. This choice is statistically grounded: both TruncatedSVD TF-IDF features and PCA-reduced BioBERT embeddings are approximately zero-centered by construction. A zero vector therefore does not simulate the presence of a document — it is an interpretable representation of *absence of semantic content*, placed at the distributional center of existing embeddings.

To prevent models from confusing a genuinely zero-valued embedding with a missing note, binary indicator variables are added for each text modality (e.g., `has_discharge_note`, `has_radiology_note`). This allows logistic regression to explicitly learn differential mortality risk associated with the presence or absence of documentation — an interpretable clinical signal in its own right, since missing notes may correlate with shorter stays, rapid deterioration, or administrative patterns.

# **THEORETICAL FRAMEWORK**

## **1. Structured Data Modelling**

Structured clinical data plays a crucial role in predicting patient outcomes. In this study, two key modelling approaches are used:

### **1.1 LASSO Logistic Regression**

LASSO (Least Absolute Shrinkage and Selection Operator) is a regularization technique applied to logistic regression that introduces an L1 penalty. This penalty shrinks some coefficients to zero, effectively performing feature selection.

- Eliminates irrelevant variables by assigning zero coefficients
- Produces a sparse and interpretable model
- Reduces overfitting
- Selected features represent statistically significant predictors

### **1.2 XGBoost (Extreme Gradient Boosting)**

XGBoost is an ensemble learning method based on gradient-boosted decision trees. It builds multiple trees sequentially to improve predictive performance.

- Captures complex and non-linear relationships
- Handles feature interactions effectively
- Provides feature importance based on contribution (gain/split importance)
- Selects top features based on predictive power

## **2. Text Data Representation (Normalization Techniques)**

Unstructured clinical text such as discharge summaries and radiology reports is transformed into numerical features using the following techniques:

## 2.1 TF-IDF (Term Frequency–Inverse Document Frequency)

TF-IDF is used to convert textual data into numerical vectors by assigning importance scores to words.

- **Term Frequency (TF):** Measures how frequently a word appears in a document
- **Inverse Document Frequency (IDF):** Measures how rare a word is across all documents

TF-IDF = TF×IDF

- Common words receive lower scores
- Rare but important words receive higher scores
- Produces high-dimensional feature space
- Dimensionality is reduced using techniques like SVD

## 2.2 BioBERT

BioBERT is a transformer-based deep learning model specifically trained on biomedical text (PubMed, PMC).

- Captures contextual and semantic meaning of medical language
- Understands domain-specific terminology and relationships
- Uses bidirectional context (left and right words)
- Provides richer feature representation compared to traditional methods

# 3. Multivariate Logistic Regression

Multivariate logistic regression is used to model binary outcomes (e.g., survival vs mortality) using multiple predictors.

### Model Formulation

$$\log \left( \frac{p}{1-p} \right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

Where:

- $p$  = probability of the event (e.g., death)
- $x_i$  = predictor variables
- $\beta_i$  = model coefficients

### **Interpretation of Coefficients**

- $\beta > 0$ : Increases probability of the event
- $\beta < 0$ : Decreases probability of the event
- Odds Ratio =  $\exp(\beta)$ 
  - $OR > 1 \rightarrow$  increased risk
  - $OR < 1 \rightarrow$  decreased risk

## **4. Integration of Structured and Unstructured Data**

The final model integrates:

- Structured clinical variables (vitals, lab values)
- Text-derived features (TF-IDF, BioBERT embeddings)

This combined approach improves predictive accuracy by leveraging both numerical and contextual information.

# ANALYSIS

## Structured Data:

### 1. Feature Selection Analysis

Feature selection was performed using both **LASSO Logistic Regression** and **XGBoost**, followed by a combined feature set.

- LASSO selected **29 features** based on statistical relationships and regularization
- XGBoost selected **25 features** based on feature importance (predictive power)
- The combined approach resulted in **42 features**, capturing both statistical significance and non-linear importance

### Interpretation

The combination of LASSO and XGBoost ensures a **balanced feature set**, where:

- LASSO contributes interpretable and statistically relevant variables
- XGBoost captures complex, non-linear interactions

This hybrid approach improves robustness and reduces the risk of missing important predictors.

### Results:

#### 1. Lasso set of features:

```
LASSO features: Index(['los', 'anchor_age', 'hr_mean', 'dbp_min', 'spo2_min',  
'dbp_max',  
    'bt_min', 'bicarbonate_mean', 'calcium_mean', 'creatinine_mean',  
    'inr_mean', 'lactate_mean', 'potassium_mean', 'wbc_mean', 'bun_min',  
    'glucose_min', 'ph_min', 'bicarbonate_max', 'hemoglobin_max',  
    'lactate_max', 'platelet_max', 'pt_max', 'copd', 'diabetes',  
    'epinephrine', 'dopamine', 'received_ventilation', 'gcs_eye',  
    'gcs_total'],
```

#### 2. XGBoost Set of features:

```
XGB features: ['gcs_verbal', 'bicarbonate_mean', 'lactate_mean', 'gcs_eye',
'bicarbonate_max', 'ph_mean', 'bun_min', 'wbc_min', 'bun_mean', 'platelet_mean',
'inr_min', 'lactate_max', 'pt_min', 'calcium_max', 'los', 'calcium_mean',
'received_ventilation', 'wbc_mean', 'pt_mean', 'glucose_mean', 'dbp_min',
'gcs_total', 'hematocrit_max', 'platelet_min', 'anion_gap_mean']
```

### 3. Combined set of features (LASSO U XGBoost):

```
glucose_mean, lactate_mean, gcs_verbal, bun_min, inr_min
anion_gap_mean, dopamine, pt_mean, pt_min, calcium_mean
wbc_min, received_ventilation, inr_mean, creatinine_mean, epinephrine
bun_mean, platelet_min, hemoglobin_max, glucose_min, wbc_mean
gcs_total, platelet_mean, pt_max, dbp_min, calcium_max
platelet_max, bt_min, lactate_max, potassium_mean, anchor_age
los, bicarbonate_max, diabetes, hr_mean, dbp_max
ph_min, spo2_min, gcs_eye, hematocrit_max, bicarbonate_mean
ph_mean, copd
```

## 2. Individual Significance Testing

Each variable was evaluated using **univariate logistic regression**, and features with **p-value < 0.05** were selected.

- Total selected features: **39**

### Interpretation

This step ensures that each predictor has a **statistically significant association with mortality**. It helps filter out irrelevant variables before building the multivariate model, improving model reliability.

## 3. Multicollinearity Assessment (VIF)

Multicollinearity was assessed using **Variance Inflation Factor (VIF)**.

- Variables with **VIF > 10** were considered highly collinear
- **8 features were removed**
- Final selected features: **31**

## Interpretation

Removing highly correlated variables:

- Reduces redundancy
- Improves model stability
- Ensures reliable coefficient estimates

This step is crucial for maintaining the **interpretability of logistic regression**.

## 4. Multivariate Logistic Regression Results

A multivariate logistic regression model was built using the final **31 features**.

### Key Findings:

- Several clinical variables showed significant association with mortality:
  - **GCS Eye** → strong predictor (neurological status)
  - **SpO2 Min** → oxygen saturation impact
  - **Epinephrine** → treatment-related variable
  - **Calcium Mean / Bicarbonate** → metabolic indicators

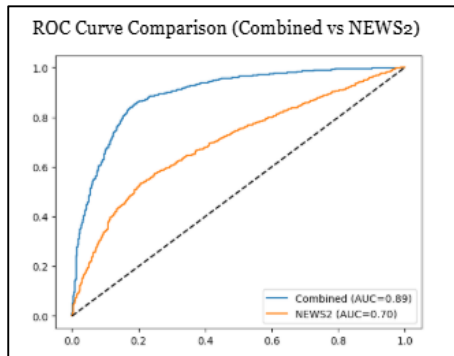
### Interpretation:

- Positive coefficients → increase mortality risk
- Negative coefficients → decrease mortality risk
- Odds ratios help quantify effect size

The model captures **independent contributions of physiological and treatment-related factors**, making it clinically meaningful.



## 5. Model Evaluation – Discrimination Performance



The model was evaluated using ROC-AUC and other metrics.

### Results:

- $AUC \approx 0.891 \rightarrow$  strong discrimination
- Accuracy  $\approx 0.825$
- F1 Score  $\approx 0.829$

- Recall  $\approx 0.814$
- Pseudo  $R^2 \approx 0.383$

### Comparison:

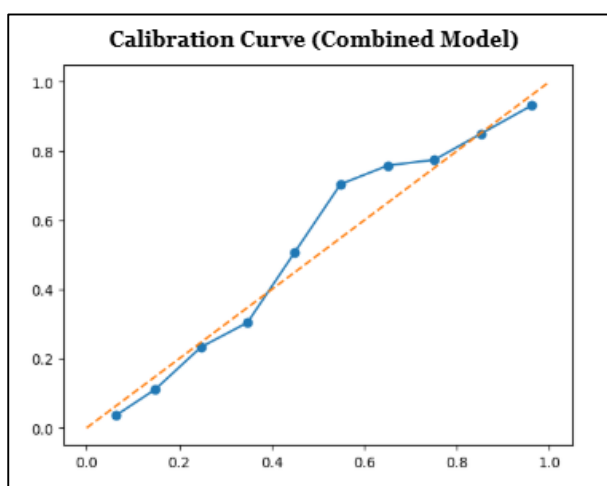
- Outperformed NEWS2 (AUC  $\approx 0.697$ )

### Interpretation:

- High AUC indicates excellent ability to distinguish between survival and mortality
- High F1 score indicates balanced precision and recall
- Higher pseudo  $R^2$  indicates better model fit

The model significantly outperforms traditional scoring systems.

## 1. Calibration Analysis



The calibration curve shows that predicted probabilities closely follow the ideal diagonal line.

### Interpretation:

- Predicted risks align well with actual outcomes
- Model is **well-calibrated**
- Provides **reliable probability estimates**

This is critical in healthcare, where probability values guide decisions.

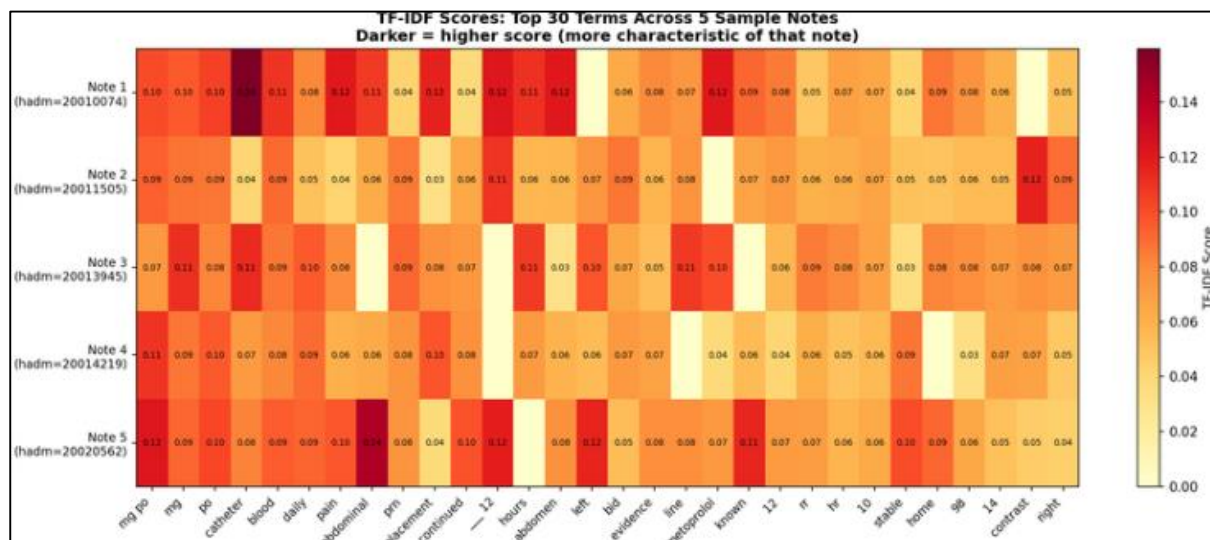
## 7. Overall Interpretation

The proposed combined model demonstrates:

- Strong **predictive performance** (high AUC, F1)
- Good **model fit** (Pseudo  $R^2$ )
- Reliable **probability estimation** (calibration)
- Effective **feature selection and stability** (VIF + LASSO + XGBoost)

## Structured + Unstructured (Text)

### 1. Text Feature Representation (TF-IDF Heatmap)



The TF-IDF heat map highlights the most important terms across clinical notes.

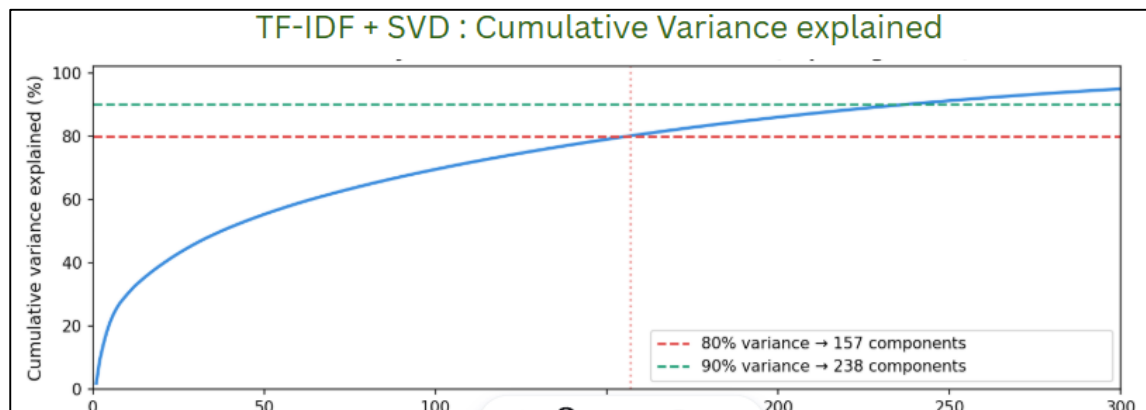
#### Interpretation

- Darker regions indicate **higher importance of specific medical terms** in individual notes
- Frequently occurring but less informative words receive lower weights
- Rare but clinically meaningful terms receive higher TF-IDF scores

This demonstrates that TF-IDF successfully captures **document-specific important keywords**, enabling meaningful representation of unstructured clinical text.

## 2. Dimensionality Reduction using SVD

The cumulative variance plot shows how much information is retained as the number of components increases.



### Key Findings

- ~80% variance explained with **157 components**
- ~90% variance explained with **238 components**

### Interpretation

- TF-IDF produces very high-dimensional data
- SVD effectively reduces dimensionality while preserving most of the information
- Selecting ~150–200 components provides a good trade-off between **efficiency and information retention**

This step improves computational efficiency and reduces noise.

## 3. BioBERT Embeddings and PCA

BioBERT embeddings convert clinical text into dense numerical vectors (768 dimensions).

## Key Observations

- Each note is represented as a **768-dimensional embedding**
- PCA reduces dimensions while preserving ~90% variance (~100–110 components)

## Interpretation

- BioBERT captures **deep contextual and semantic meaning**
- PCA helps in reducing redundancy and computational complexity
- Compared to TF-IDF, BioBERT provides **richer and more meaningful representations**

# 4. Multimodal Feature Selection (Structured + Text)

## LASSO Results

- Selected **64 non-zero features**
- Includes both structured variables and text-derived features

## XGBoost Results

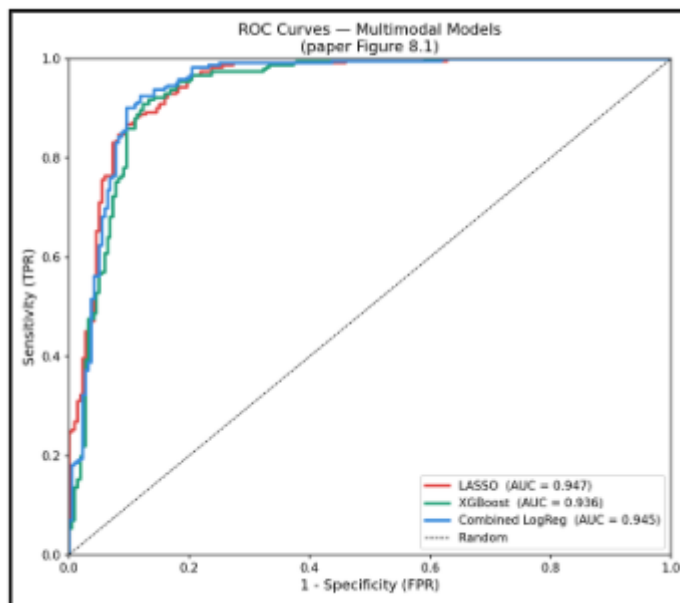
- Selected top **64 important features**
- ~30 features overlap with LASSO

## Interpretation

- Both models identify **key predictors consistently**
- Overlapping features indicate **robust and reliable variables**
- Important predictors include:
  - Clinical variables (GCS, HR, lactate)
  - Text-based features (TF-IDF, BioBERT embeddings)

This confirms the importance of **multimodal integration**

## 5. Model Performance – ROC Analysis



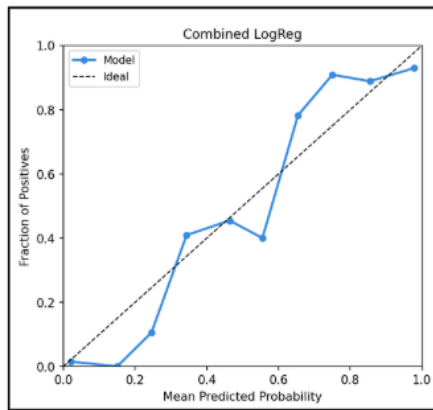
### Results

- LASSO AUC  $\approx 0.947$
- XGBoost AUC  $\approx 0.936$
- Combined Logistic Regression AUC  $\approx 0.945$

### Interpretation

- All models show **excellent discrimination ability**
- Curves are close to the top-left corner  $\rightarrow$  strong classification performance
- Minimal difference between models indicates **stable performance across methods**

## 6. Calibration and Clinical Usefulness



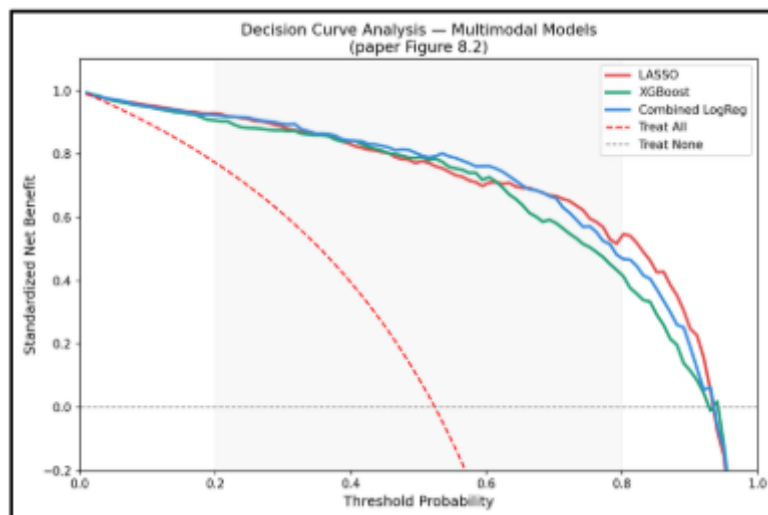
## Calibration Curve

- Closely follows diagonal line

Interpretation:

- Predicted probabilities match actual outcomes
- Model is **well-calibrated and reliable**

## Decision Curve Analysis



- Model curves lie above:
  - Treat All
  - Treat None

Interpretation:

- Model provides **positive net benefit**
- Useful for real-world clinical decision-making
- Avoids unnecessary interventions while identifying high-risk patients

## Comparative Analysis (Structured vs Multimodal)

### Structured Only Model

- AUC  $\approx 0.891$
- F1  $\approx 0.829$
- Recall  $\approx 0.814$

### Structured + Text Model

- AUC  $\approx 0.944\text{--}0.945$
- F1  $\approx 0.892$
- Recall  $\approx 0.921$

### Interpretation

- Significant improvement after adding text data
- Higher recall  $\rightarrow$  better detection of high-risk patients
- Higher F1 score  $\rightarrow$  balanced performance

Multimodal model clearly **outperforms traditional structured-only models**

## **CONCLUSION**

This project developed and evaluated a multimodal predictive framework for in-hospital mortality among cardiac arrest ICU patients using the MIMIC-IV dataset. By integrating structured clinical data (vital signs, laboratory values, comorbidities, treatment indicators, and neurological assessments) with unstructured clinical text (discharge summaries and radiology reports processed via TF-IDF and BioBERT embeddings), the study demonstrated substantial improvements in predictive performance over structured-only and traditional scoring models.

The structured combined model achieved an AUC of 0.89 and outperformed the NEWS2 clinical scoring system (AUC = 0.70) by a significant margin. The multimodal combined model further improved performance to an AUC of approximately 0.945, with corresponding improvements in accuracy, F1-score, and recall. The higher Pseudo- $R^2$  (0.64 vs. 0.38) of the multimodal model confirms that unstructured clinical text captures clinically meaningful information not present in structured records alone.

The findings highlight the potential of multimodal EHR modeling to improve mortality risk prediction and support clinical decision-making. Although the analysis was based on data from a single institution and post hoc notes, the framework can be readily extended to other hospitals and real-time contexts. Future work should focus on temporal modeling of early clinical narratives, external validation across healthcare systems, and improving interpretability of text-derived features. Overall, this study provides empirical evidence that integrating structured and unstructured data enhances predictive performance and clinical insight, supporting the development of more accurate and explainable decision-support tools in critical care.



## **FUTURE SCOPE**

1. **Multi-Center Validation:** Testing the model on external datasets (such as ICU or local hospital databases) to ensure the predictive power remains consistent across different healthcare systems with varying patient demographics and clinical documentation styles.
2. **Long-Sequence Transformers:** Implementing newer transformer architectures (such as BigBird or Longformer) that can process entire clinical notes without truncation, ensuring that no clinically relevant medical context is lost due to token limitations.
3. **Real-Time Integration:** Developing a live dashboard version of the model that updates mortality risk scores in real-time as new laboratory results, vital signs, and nursing notes are entered into the electronic health record system, enabling proactive clinical intervention.
4. **Prospective Clinical Trials:** Moving from retrospective data analysis to prospective shadow testing in a real ICU environment to evaluate clinical utility, usability, and physician trust in the model's predictions.
5. **Expanding Modalities:** Incorporating more diverse data types, such as raw ECG waveforms or medical imaging (X-rays and CT scans), directly into the meta-model to provide a more holistic and comprehensive view of patient health status.

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### **Appendix:**

Drive Link for codes: [Source codes of Overall Research](#).